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Individual Project, 2025 - 2026 Instruction Manual

Introduction (Requirements / Short Description)

This application is intended for users who want to manage their daily tasks while staying motivated through gamification elements such as levels, ranks, streaks, and achievements. The main purpose of the application is to improve productivity by transforming task management into an engaging and rewarding experience.

The primary functionalities of the application include:

- **User management with roles** – The system supports different user roles (ADMIN and PLAYER), allowing administrators to manage users while players focus on completing tasks.
- **Task creation and management** – Users can create, edit, delete, and complete tasks with different types, difficulties, and categories.
- **Gamification system** – Users earn points, increase their level, maintain streaks, unlock achievements, and progress through ranks based on their activity.
- **Persistence and data storage** – Application data is stored either in text files or in an Oracle SQL database, ensuring data persistence between sessions.

By using this application, users benefit from increased motivation, better organization of tasks, and clear visual feedback on their progress and achievements.

Design

The application architecture is based on the following technologies:

- **Backend:** Java (Object-Oriented Programming, JDBC for database access)
- **Frontend:** Command-Line Interface (CLI), with planned extension to JavaFX GUI
- **Database:** Oracle SQL (USERS, TASKS, ACHIEVEMENTS tables)
- **Persistence Layer:** File-based storage (TXT files) and database storage (Oracle)

The system follows a layered architecture, separating domain logic, service logic, and persistence logic to allow future extensions such as a graphical user interface or a web-based version

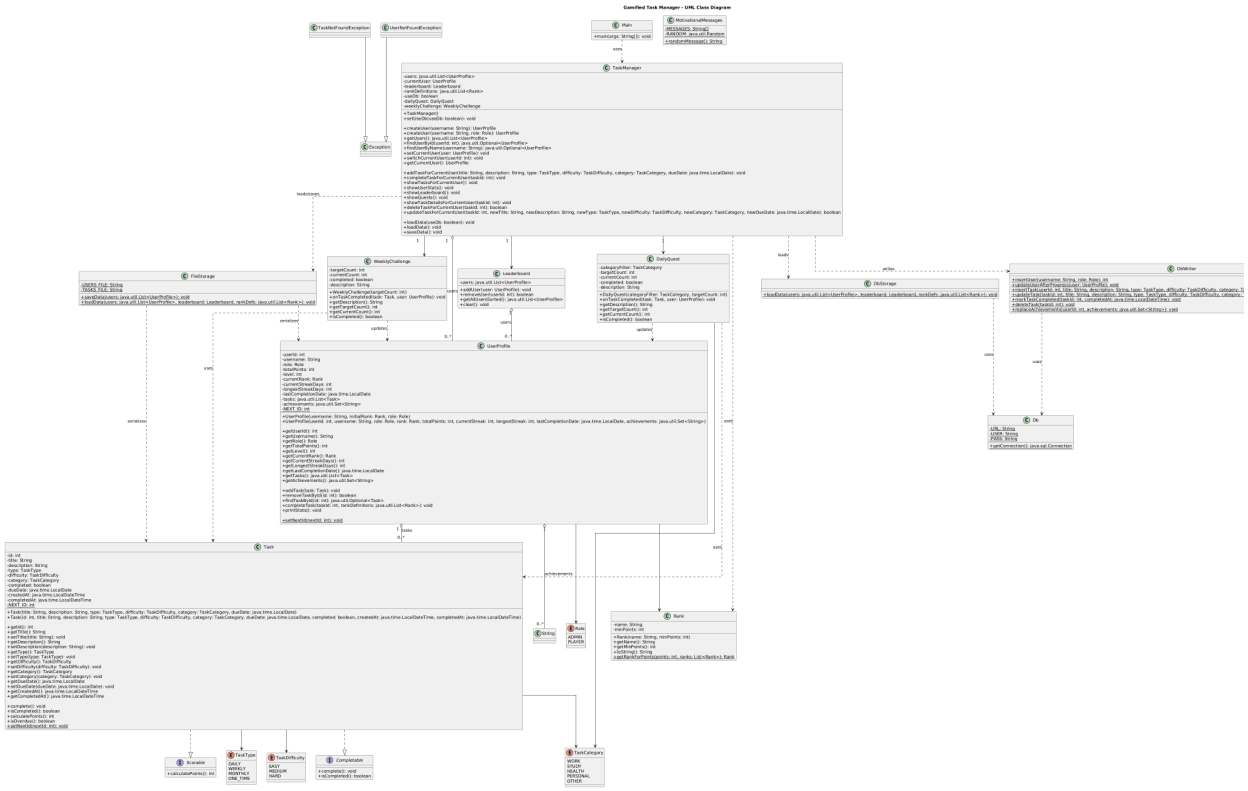
Diagrams

The application uses UML diagrams to describe its structure and behavior.

The **Class Diagram** illustrates the main components of the system, including:

- `TaskManager` as the central controller of the application
- Domain models such as `UserProfile`, `Task`, and `Rank`
- Gamification components like `DailyQuest` and `WeeklyChallenge`
- Persistence components such as `FileStorage`, `DbStorage`, and `DbWriter`
- Relationships between users, tasks, achievements, and ranks

The UML diagram clearly shows one-to-many relationships between users and tasks, as well as between users and achievements, ensuring a clear and well-structured design



Implementation

`To continue the development of the application from the provided zip folder for the whole project, the client can follow these steps:`

1. **Navigate to the project directory:**

```
cd gamified-task-manager
```

2. **Compile the project:**

```
mvn clean install
```

3. **Run the application:**

```
java -jar target/gamified-task-manager.jar
```

The application can be started in either file-based mode or database mode, depending on the configuration.

(Ensure that the Oracle database container is running before starting the application.)

Usage

Below are examples of how the application can be used and how users interact with its features.

1. Authentication and User Roles

Users can log in using their username and password. Administrators have additional privileges such as creating users, deleting users, and resetting player statistics, while players focus on task completion.

2. Interface Navigation

The application presents a menu-based interface that dynamically adapts based on the user's role. Admin users see administrative options, while player users see only task-related options.

3. Using Main Functions

The application offers several essential functionalities for users:

○ **Creation and management of tasks** – Users can add, edit, delete, and complete tasks. Tasks can be categorized, assigned a difficulty level, and filtered based on different criteria.

○ **Gamification and progress tracking** – Completing tasks rewards points, increases levels, updates streaks, unlocks achievements, and promotes users through ranks such as Bronze, Silver, and Gold.

○ **Data persistence** – User data, tasks, and achievements are stored in a database or text files, allowing users to resume their progress across sessions.

○ **Leaderboard and challenges** – Users can view a leaderboard comparing progress and participate in daily quests and weekly challenges.

4. Usage Examples

Typical usage scenarios include:

○ Adding a new task and completing it to gain points.
○ Viewing completed, pending, and overdue tasks.
○ Unlocking achievements after reaching milestones.
○ Comparing progress with other users on the leaderboard.

Conclusions and Next Steps

This application provides a robust and engaging solution for task management through gamification. Users benefit from increased motivation, structured task organization, and clear progress tracking.

Future improvements may include:

- Adding a modern graphical user interface using JavaFX.
- Introducing additional gamification elements such as mystery boxes and rewards.
- Expanding the system into a web-based application.
- Improving performance and scalability for a larger number of users.